CATEGORY B

Coding Assignment

Problem 2: Cancelled Orders Cleanup

Difficulty: Medium · **Estimated time:** 25–30 minutes

TASK

A delivery app keeps order codes in a singly linked list. When a customer cancels, every order with that code must be removed.

Implement `Node removeAll(Node head, int code)` that removes every node with the given value and returns the (possibly new) head. Write your own node class with `data` and `next` fields; do not use `java.util.LinkedList`.

INPUT / OUTPUT

Input: A list and the code to remove.

Output: The list after removal.

CONSTRAINTS

- 0 <= length of list <= 1000

EXAMPLES

Sample 1

```
orders = 5 -> 3 -> 5 -> 8 -> 5, code = 5

Expected Output: 3 -> 8 -> null
```

Sample 2

```
orders = 5 -> 5 -> 5, code = 5

Expected Output: null
```

COMPLEXITY REQUIREMENTS

State the time and additional space complexity.

HINT

First remove matching nodes from the front by moving head; then walk with a previous reference and skip matching nodes.